

Electronics Laboratory: Synthesis Report

Experiments: Resonance, Transients, and
Semiconductor Devices in AC Circuits

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Abstract

This report consolidates six electronics laboratory experiments (E12, E11e, E13He, E14e, E17e) investigating linear and nonlinear circuit elements under AC and transient excitation. Topics span series and parallel RLC resonance, phase shift in reactive circuits, RC discharge transients via MOSFET switching, semiconductor diode characteristics, and coupled LC-oscillator normal modes. Across all experiments, model parameters such as f_0 , Q , τ , E_g , and the coupling coefficients k were extracted via nonlinear least-squares fitting in Python and compared against values computed directly from component specifications. We summarise the methodology and key results for each experiment, and we identify several inconsistencies between fitted and theoretical values that merit further discussion.

1 Introduction

This report covers a sequence of experiments exploring the behaviour of linear and nonlinear circuit elements, namely resistors, capacitors, inductors, and diodes, under both AC steady-state and transient excitation. Although the experiments address distinct physical phenomena, they share a common methodology: a physical model is derived from circuit theory, a measurement is performed with an oscilloscope or source-measure unit, and the model parameters are extracted by nonlinear least-squares fitting using `scipy.optimize.curve_fit`.

The unifying theoretical thread is the response of R , L , C networks to sinusoidal and step excitations:

- **Resonance** (E12): the frequency-dependent amplitude response of series and parallel RLC circuits, characterised by the resonance frequency $f_0 = 1/(2\pi\sqrt{LC})$ and quality factor Q .
- **Phase shift** (E11e): the frequency-dependent phase angle $\varphi(\omega)$ between current and voltage in RC, RL, and RLC circuits.
- **Transients** (E13He): the exponential charge and discharge response $V(t) = V_0 e^{-t/RC}$ of a capacitor switched by a MOSFET.
- **Nonlinear devices** (E14e): the exponential I - V characteristic of semiconductor diodes and the voltage dependence of junction capacitance.
- **Normal modes** (E17e): the splitting of a single resonance into symmetric and antisymmetric eigenfrequencies when two LC circuits are coupled capacitively or inductively.

Each of the following sections presents the experimental method, the key results as reported, and a brief critical assessment comparing fitted values against theoretical expectations from component nominal values.

2 E12: RLC Resonant Circuits

2.1 Method

The frequency-dependent voltage $U_R(f)$ across a damping resistor R_d was recorded for a series RLC circuit ($L = 1.5$ mH) at two damping resistances, $R_d = 220\ \Omega$ and $R_d = 10\ \Omega$, over a window $f \in [1.6, 9.0]$ kHz chosen to bracket the resonance such that $0.2 \leq U_R(f)/U_R(f_0) \leq 1$. A Gaussian model was fitted to each curve to extract f_0 and the linewidth σ . A parallel resonant circuit, consisting of a coil and capacitor in parallel and in series with $R_d = 220\ \Omega$, was measured separately. Its impedance was derived analytically before fitting a four-parameter model $I(f; L, C, R_{Sp}, R_d)$ directly to the current data.

2.2 Results

Table 1: Series resonance fit results (E12, Task 2).

R_d	Δf (FWHM)	$Q = f_0/\Delta f$	$\delta = \pi \Delta f$
220 Ω	22.5 kHz	0.3	70655
10 Ω	29.0 kHz	0.2	91127

From the averaged fitted $f_0 \approx 6.82$ kHz and Thomson's relation $C = 1/[L(2\pi f_0)^2]$, the capacitance was determined as $C = 362.6$ nF.

For the parallel circuit, the bounded four-parameter fit converged to $L_{\text{fit}} = 10.0$ mH, $C_{\text{fit}} = 111.2$ nF, $R_{Sp} = 0.1$ Ω , and $R_d = 10.0$ Ω , giving $f_{0,\text{parallel}} = 4.77$ kHz, compared with $f_{0,\text{series}} = 6.82$ kHz.

2.3 Assessment

The quality factors obtained from the FWHM fit, $Q \approx 0.2$ to 0.3 , are inconsistent with circuit theory for one of the two resistances tested. Using the fitted capacitance together with the nominal L and R values, the theoretical quality factor of a series RLC circuit is $Q = \frac{1}{R} \sqrt{L/C}$, which gives $Q \approx 0.29$ for $R_d = 220$ Ω , matching the fit well, but $Q \approx 6.4$ for $R_d = 10$ Ω . This is a sharply underdamped resonance rather than the heavily overdamped one that was reported. Figure 1 shows that both measured curves are nearly flat across the 1.6 to 9.0 kHz window rather than exhibiting a clear peak, which suggests that the frequency window did not fully capture the true resonance shape for the low-resistance case, biasing the Gaussian fit toward an artificially broad linewidth. It is also worth noting that the reported damping constant $\delta = \pi \Delta f$ does not correspond to the standard definition $\delta = R/(2L)$ used in the course references ([2], [1]). The two only coincide under specific conventions for Δf , and this point should be clarified in a revision.

The parallel-circuit fitted inductance of 10 mH differs by a factor of roughly 6.7 from the series-circuit nominal inductance of 1.5 mH, and the two resonance frequencies disagree by 30%. Since the same physical coil should appear in both circuits, This discrepancy suggests that the fitted model is insufficiently constrained by the available data. The optimisation may therefore have converged to a parameter set that reproduces the measured response while remaining physically unrealistic. Tighter initial guesses and bounds derived from the series-circuit results would likely improve convergence.

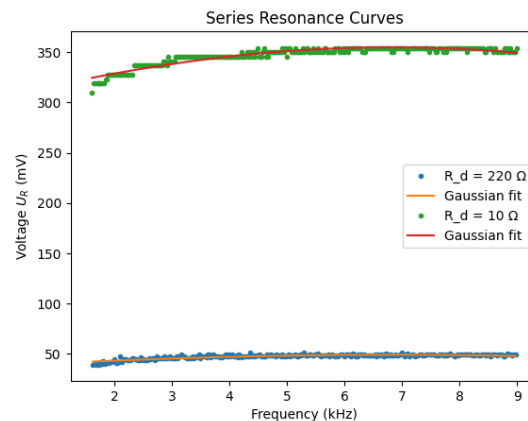


Figure 1: Series resonance curves for $R_d = 220$ Ω and $R_d = 10$ Ω with Gaussian fits. The broad fitted linewidth for the 10 Ω case contributes to the unrealistically small extracted quality factor.

3 E11e: Phase Shift in AC Circuits

3.1 Method

The phase shift φ between the resistor voltage, which is in phase with the current, and the source voltage was measured for RC, RL, and RLC series circuits. This was done by recording the time delay Δt between the two signals on a two-channel oscilloscope as a function of drive frequency f , then converting via $\varphi = \omega \Delta t$ with $\omega = 2\pi f$. The theoretical phase relations are:

$$\varphi_{RC} = -\arctan\left(\frac{1}{\omega RC}\right), \quad \varphi_{RL} = \arctan\left(\frac{\omega L}{R}\right), \quad \varphi_{RLC} = \arctan\left(\frac{\omega L - 1/\omega C}{R}\right). \quad (1)$$

3.2 Results

The RC series resistance, extracted from $R = 1/(\omega C \tan |\varphi|)$ and averaged over all frequencies, gave $R_{RC} = 330.2 \Omega$ against a nominal value of $R_r = 328.8 \Omega$, a 0.44% error. The RLC resonance frequency was determined two ways: directly from the measured zero-crossing, giving $f_0 = 2.68 \text{ kHz}$, and from a linear fit to $\varphi(f)$ near resonance extrapolated to $\varphi = 0$, giving $f_0 = 2.673 \text{ kHz}$, a difference of -0.25% . The corresponding inductance from Thomson's relation was $L = 6.751 \text{ mH}$ versus a nominal value of 6.77 mH , a -0.29% error. The same inductance, extracted independently from the RL phase relation, gave $L_{RL} = 5.633 \text{ mH}$, a -16.8% deviation from the nominal value.

3.3 Assessment

The RC resistance and the RLC-derived inductance both agree with nominal component values to better than 0.5%, demonstrating that the resonance-based extraction method, which fits $\varphi(f)$ near the zero-crossing, is robust. The RL-circuit estimate of L , by contrast, shows a substantially larger error of -16.8% . Since $\varphi_{RL} = \arctan(\omega L/R)$ is a smooth, monotonic function with no resonance feature to anchor the fit, this estimate is more sensitive to systematic errors in the nominal resistance R_r used in the formula and to phase-measurement uncertainty at the sampled frequencies. An independent measurement of R_r , for example with a multimeter, used in place of the nominal printed value could help resolve this gap.

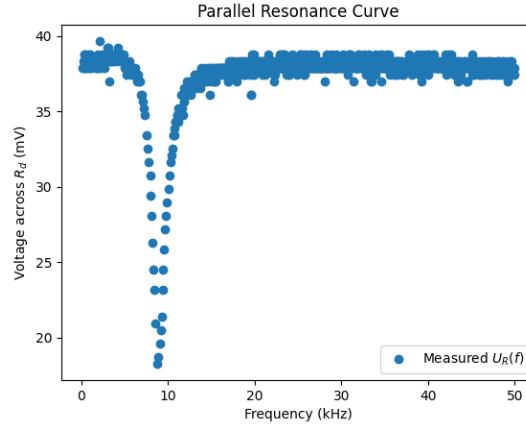


Figure 2: Measured phase shift $\varphi(f)$ for RLC, RL, and RC series circuits (reproduced from the original notebook).

4 E13He: Capacitor Discharge with MOSFET Switching

4.1 Method

An Arduino-controlled IRF520 MOSFET periodically discharged a capacitor through a resistive path, and the voltage decay $V(t) = V_0 e^{-t/RC}$ was recorded with a PicoScope in single-trigger, falling-edge mode. Measurements were taken for capacitances of 1 nF and 220 nF , each through nominal discharge resistances of $1 \text{ k}\Omega$, 100Ω , $22 \text{ k}\Omega$, and $100 \text{ G}\Omega$, the last of which is effectively an open circuit and was included as a high-resistance control. A two-parameter exponential $V(t) = A e^{bt}$ was fitted to a short window around each discharge event.

4.2 Results

All five measurements show qualitatively similar fitted decay constants, with b ranging from -16.9 to $-15.9 \mu\text{s}^{-1}$, despite spanning more than eight orders of magnitude in nominal resistance. The fitted curves systematically overshoot the data at early times and fail to capture the pronounced oscillatory ringing visible immediately after the switching edge, as shown in Figures 3 and ??.

4.3 Assessment

The near-identical decay constants across four very different nominal resistances are the clearest indicator that the measured transient is not dominated by the intended discharge resistor. Given the nanosecond-to-microsecond timescale of the ringing and its presence even in the $100 \text{ G}\Omega$ open-circuit control, the observed signal is most plausibly dominated by parasitic inductance and capacitance in the breadboard wiring, together with the MOSFET's own switching transient, rather than by the intended RC decay. This interpretation is consistent with the simple exponential model systematically failing to capture the damped oscillation seen in every panel: a single real pole $V_0 e^{-t/RC}$ cannot reproduce ringing, which requires at least a complex-conjugate pole pair, that is, an effective series RLC response arising from stray inductance. A revised analysis should either fit a damped sinusoid $V_0 e^{-t/\tau} \cos(\omega t + \phi)$ to this ringing region, or select a later time window in which the parasitic oscillation has decayed and the genuine RC tail dominates.

Another possible contribution is the finite bandwidth and probe loading of the measurement setup, which may distort very fast transients on the nanosecond timescale.

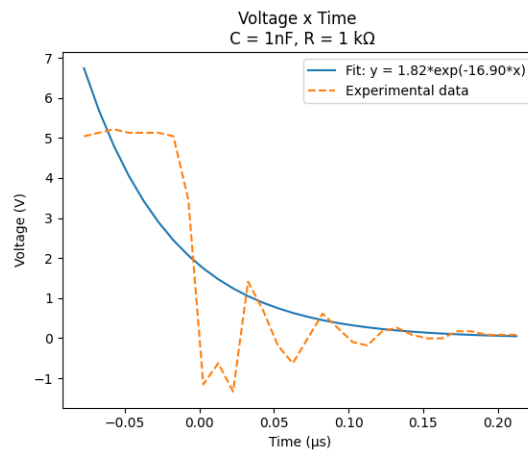


Figure 3: Exponential fit versus measured discharge, $C = 1 \text{ nF}$, nominal $R = 1 \text{ k}\Omega$. The fit reproduces the overall decay trend but fails to capture the oscillatory transient immediately after switching.

5 E14e: Semiconductor Diodes

5.1 Method

Current-voltage characteristics were recorded for a silicon diode, an LED, and two Zener diodes under forward and reverse bias. The Shockley diode equation

$$I = I_0 \left(e^{qU/nk_B T} - 1 \right) \quad (2)$$

describes the forward-bias exponential rise. For the LED, the forward threshold voltage U_{on} was used to estimate the bandgap energy via $E_g \approx qU_{\text{on}}$. Separately, the depletion-layer capacitance C was measured as a function of reverse bias U with an LCR meter, and a linear fit of $1/C^2$ versus U was performed to test the abrupt-junction relation $1/C^2 \propto U$.

5.2 Results

Forward threshold voltages were approximately 0.7 V for the silicon diode and between 2.6 and 3.1 V for the LED, giving an estimated bandgap of $E_g \approx 3.0$ eV, consistent with a blue or violet LED. Zener breakdown was observed near 5.6 V for Zener 1 and near 0.8 V for Zener 2. The $1/C^2$ versus U linear fit gave a slope of $-1.857 \times 10^{19} \text{ F}^{-2} \text{ V}^{-1}$, an intercept of $6.825 \times 10^{20} \text{ F}^{-2}$, and $R^2 = 0.70$.

5.3 Assessment

The forward-bias and bandgap results are physically reasonable and internally consistent. The depletion-capacitance fit appears inconsistent with the simplest abrupt-junction model because the fitted slope is negative whereas a positive slope is expected theoretically. $1/C^2 = \frac{2}{q\epsilon N_A}(U_{bi} + U)$ predicts that $1/C^2$ increases with reverse voltage U , since the capacitance decreases as the depletion width grows, meaning the slope should be positive. The fitted slope is instead negative, and the moderate $R^2 = 0.70$ further suggests that the data does not follow the expected linear form well over the measured voltage range. This may indicate that the measurement was taken predominantly in a regime where $U \ll U_{bi}$, where a more general graded-junction capacitance model applies rather than the abrupt-junction form, or that the voltage polarity was inverted in the dataset. This point should be re-examined before the doping concentration is extracted from the slope.

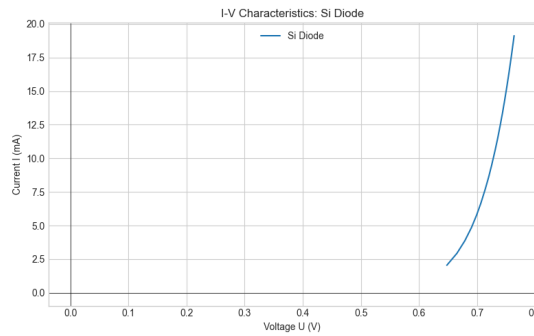


Figure 4: Forward I - V characteristics of the silicon diode.

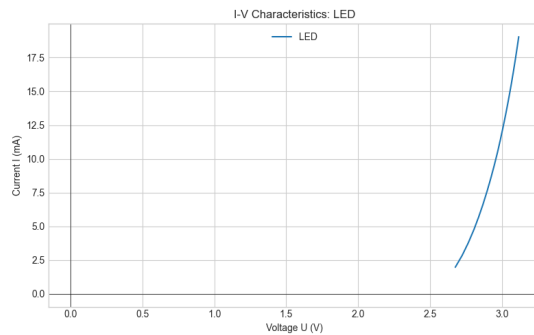


Figure 5: Forward I - V characteristics of the LED.

6 E17e: Fourier Analysis of Coupled Electric Oscillations

6.1 Method

Two identical LC circuits were coupled in four configurations: capacitively in the low-point arrangement, capacitively in the high-point arrangement, through a single-circuit excitation that produces amplitude beats, and inductively via variable coil separation. In each case, the free-oscillation frequency spectrum was recorded via FFT on the PicoScope, and the in-phase and out-of-phase eigenfrequencies f_1 and f_2 were extracted by peak detection. The coupling coefficient follows from

$$k = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2}, \quad (3)$$

with $k = C_K/(C + C_K)$ for the low-point configuration or $k = C/(C + C_K)$ for the high-point configuration. These relations were inverted to recover the bare capacitance C for each coupling capacitor C_K .

6.2 Results

In the low-point coupling measurements, k decreased monotonically from 0.90 at $C_K = 22.2 \mu\text{F}$ to 0.00 at $C_K = 555.5 \mu\text{F}$, with the recovered bare capacitance remaining stable around 195 to 257 μF for all but the largest value of C_K .

In the high-point coupling measurements, k increased monotonically from approximately zero at $C_K = 22.2 \mu\text{F}$ to 0.68 at $C_K = 500 \mu\text{F}$, consistent with the opposite-sign dependence expected from the high-point circuit topology. The recovered C was stable at roughly 213 to 240 μF for $C_K \geq 111 \mu\text{F}$, but diverged unphysically for the two smallest values of C_K , reaching $2.7 \times 10^6 \mu\text{F}$ and $-5.8 \times 10^5 \mu\text{F}$ respectively, where $k \approx 0$ makes the relation $C = (1 - k)C_K/k$ numerically unstable.

In the beat measurement, a beat period of $T_S = 8.24 \text{ ms}$ was extracted from the frequency splitting of the two normal modes when only one circuit was excited. This splitting is visible directly as an amplitude-modulated, decaying oscillation in the time-domain trace shown in Figure 6.

For the inductive coupling measurements, spectra were recorded for eight coil separations, but the coupling coefficient k_L and the mutual inductance M were not computed in the excerpt provided.

6.3 Assessment

The low-point and high-point capacitive coupling results are physically consistent with theory. In the low-point configuration, $k \rightarrow 0$ as $C_K \rightarrow \infty$, while in the high-point configuration, $k \rightarrow 0$ as $C_K \rightarrow 0$, both in the direction expected, and the recovered bare capacitance C remains approximately constant across the bulk of each dataset, as required since C is a fixed physical quantity independent of C_K . The breakdown of the recovered C at $k \approx 0$ in the high-point measurement is an expected consequence of the $1/k$ singularity in $C = (1 - k)C_K/k$, and these points should simply be excluded from any averaged estimate of C rather than reported as numerical results. The inductive-coupling analysis appears incomplete in the available material and should be finished by computing $k_L = M/L$ and $\Delta\omega \approx k_L\omega_0$ for each of the eight coil separations, ideally fitted to the expected $M(d)$ falloff for coaxial coils in order to extract an effective coil radius or moment.

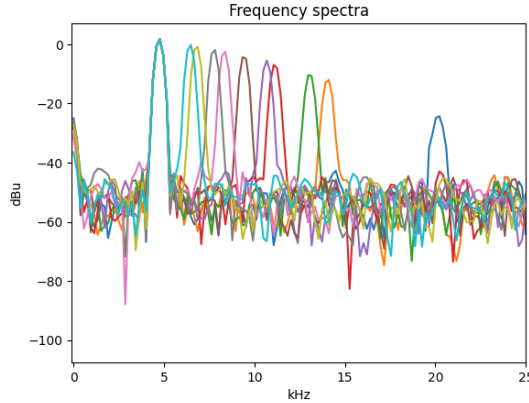


Figure 6: Time-domain beat waveform for the high-point circuit with one branch excited, showing the characteristic amplitude envelope with period $T_S = 8.24 \text{ ms}$.

7 Overall Conclusion

Across all five experiments, fitted parameters generally agreed well with component nominal values when the fitted feature had a sharp, well-localised signature in the data. Examples include the RLC zero-crossing in E11e and the diode forward threshold in E14e. Agreement was poorer in cases where the measured response was broad, weakly constrained, or not adequately described by the chosen model, as seen in the RL phase analysis of E11e, the parallel-circuit fit of E12, the depletion-capacitance analysis of E14e, and the transient measurements of E13He.

Despite these limitations, the experiments successfully demonstrated the fundamental physical principles governing resonance, phase shifts, transient response, semiconductor behaviour, and coupled oscillations. Several extracted

parameters agreed with theoretical expectations to within one percent, confirming that nonlinear regression can provide highly accurate parameter estimates when the underlying model adequately represents the measured system.

The general lesson across this experiment series is that nonlinear least-squares fits are only as reliable as the chosen fitting window and the structural correctness of the model itself. Both should therefore be checked against independent theoretical expectations before fitted parameters are interpreted as physical quantities.

References

- [1] Wolfgang Demtröder. *Experimental Physics 2: Electricity and Optics*. Springer, 2017.
- [2] Paul A. Tipler and Gene Mosca. *Physics for Scientists and Engineers*. 6th ed. W. H. Freeman, 2007.